

The Missing Piece in a Problem Facing the Curse of Dimensionality

A critical commentary of Martin and Nagel (2022):
Market Efficiency in the Age of Big Data

Nico Rosamilia[§] Camilla Sorrentino[‡]
Entrepreneurship and Entrepreneurial Finance
Politecnico di Milano - August, 2022

Abstract

The adoption of advanced econometrics techniques and machine learning algorithms receives greater attention every year also in finance. In order to extend the use of these models in entrepreneurship and entrepreneurial finance, we comment on a paper that studies a high-dimensional prediction problem for market efficiency. The rest of the work examines how and why we selected the paper. The first section also emphasizes how the authors fill the gap in the literature. Finally, we show the theoretical lens used and why we believe they are appropriate. The second section focuses on the empirical methods used. In this section, we highlight the main shortcoming of the paper while also stressing where Martin and Nagel (2022) depart from methods used by other papers addressing similar topics. The commentary ends with conclusions and future research directions.

Keywords: Bayesian learning, High-dimensional prediction problems, Return predictability, Out-of-sample tests.

[§]Email: nico.rosamilia@polimi.it

[‡]Email: camilla.sorrentino@polimi.it

1 Motivation

The choice of the paper was mainly driven by our desire to interconnect topics addressed in the course with our interest in machine learning. As discussed in class, it seems plausible to believe that these advanced computational techniques will soon be adopted in entrepreneurship and entrepreneurial finance. Indeed, we conclude our commentary with possible extensions and applications in this field.

As the title of the paper suggests, the authors study a topic that we have extensively discussed in class: market efficiency. The lens adopted in the course focuses on the agency theory and the information availability and acquisition by individuals. Regarding market efficiency, Martin and Nagel (2022) highlight and advance the literature considering a high-dimensional prediction problem to reassess the conventional wisdom on market efficiency. The second layer of connection matches the behavioral theories of entrepreneurial entry (i.e., the theory of reasoned action and the theory of planned behavior) with Bayesian updating information. Indeed, we believe that the latter represents one of the best methods to model the processing of information that underlies attitudes and subjective norms. Therefore, we believe that the lens through which market efficiency and information acquisition are used by Martin and Nagel (2022) is appropriate and fills an important gap in this strand of literature. Additionally, it models a behavioral aspect that interests the behavioral theories of entrepreneurial entry (but also for individuals' information acquisition in entrepreneurial finance as we will discuss later).

2 The Critique

2.1 The Heterogeneity and the Gap Between Model and Empirical Setting

Where does outweighing the curse of dimensionality stop? The missing piece of the paper seems to be the coexistence of a high-dimensional set of variables and the non-linearity fitting assumption (note that considering a higher-order polynomial does not mean departing from linearity). Indeed, while the model stresses the importance of dealing with high dimensional sets, the empirical application introduces nonlinearity but reduces the number of features (it considers only historical returns). We move two main critiques to the paper.

Firstly we believe that the paper may have introduced heterogeneity across individuals. Indeed, an immediate source of heterogeneity should consider that agents have different predictors J (the set of variables predicting prices). Additionally, they have different info availability and, lastly, knowledge. These considerations harness the paper into similar shortcomings of the market efficiency theory under rational expectation. These concerns hold true not only with respect to perfect predictability and forecasting errors but mainly in comparison with the relation between returns exogeneity and individuals' profit realization. In other words, we believe that addressing heterogeneity across individuals would add relevant features to the performances of the market and the behavior of the agents. Indeed, we find it relevant to introduce heterogeneity in this frame because it encompasses two layers of originality and novelty: the first one allows researchers to increase the variance of the features for price prediction (adding to the Fama and French (1993) five factors model, which seems to be one of the goals of the authors, see the model in chapter two; the second layer, in our opinion, may finally shed light to the strand of literature supporting the idea that financial markets are a zero-sum game. Indeed, we believe that individual heterogeneity is definitely one of the drivers of symmetric gains in financial investing. For clarification, see the example in the appendix A.1.

Additionally, it would have been interesting to shed light on the behavioral component that considers individuals not only as a single being but already as Bayesian thinkers that are also able to act in conjunction with other individuals (or unconsciously influenced by others - an extreme example: the Game Stop case). Indeed, the authors slightly address this point in the conclusion and suggest directions for further research. This argument may add to the herding behavior school of thoughts and simultaneously account for bounded rationality and behavioral heterogeneity of individuals. Finally, we think that the authors do not fully convince us with the explanation regarding the choice of J predicting factors as being constant over time. Similarly, they do not justify at all the assumption of zero interest rate (we believe that it simplifies the study when Martin and Nagel (2022) think of a period as a long time span and therefore departing from discounting and compounding for cash flows in different time periods).

The second major point we would like to highlight involves the absence of a wedge between the model and the empirical setting. Indeed, the authors do not test empirically the model developed in order to highlight the most important predicting variables. Instead, they use past prices that, by nature lagged variables, do not exactly fit into their model for pricing (the matrix of predictable variables X should not include lagged variables). The authors force the link with the model arguing that the J predictors may be lagged variables at a different level of persistency). Additionally, in our view, it appears that they do not make the case to explain why the empirical setting departs from the model. This may be a crucial point that leaves the reader with a hole in the storyline. Conversely, Martin and Nagel (2022) strongly motivate

their choice by focusing on the creation of the wedge between in-sample and out-of-sample predictability¹.

2.2 A Discussion on the Empirical Method

In chapter 7, Martin and Nagel (2022) deploy the ridge regression not specifying why the choice of a second-order polynomial (they say that they used squared lagged returns. Ergo we think that there should be a clearer motivation for such a fitting choice - indeed we may be pushed to believe that the wedge between in-sample and out-of-sample predictions may also derive from the data set fitting choice. Machine learning techniques come in handy in these cases). In this regard, in addition to the wedge between the two samples, we were expecting a stronger motivation for the choice of the model for variable importance. Indeed, from the literature emerges that random forests and extremely randomized trees are more robust for classification, prediction, and forecasting problems (see Michalski and Low (2021)). We believe that making the case for a model overlapping and combining Bayesian updating information and machine learning represents a step ahead for the behavioral model robustness. Furthermore, the motivation of empirically testing the price history seems to be missing and it may also bear a choice bias. Indeed, while the model proposes to study a set of predictors (that may actually be higher than the number of observations $J > N$), the empirical analysis uses a common subset of variables with known predicting power (the authors acknowledge this issue when referring to momentum and long-run reversal. However, they do not seem to motivate enough the choice with respect to the underpinnings built with the model). Moreover, a small comment on the choice of the ridge regression was expected. Indeed, one may argue that a machine learning prediction algorithm overcomes the weight choice issue, automatically selecting the most relevant variables. Finally, the authors superficially address the shift in the emphasis on a different time window for returns. This may have been tested in the model clustering lagged returns in different time frames in order to test which one is more significant in their analysis and therefore contribute to the strand of literature concerning momentum. Martin and Nagel (2022) include lags up to a certain point (it would have been nice a justification for the choice of this point. We guess that it was chosen the farthest rolling window time frame considered in the literature). Therefore, the data snooping problem does not persist within the choice of these specifications but, in our opinion, it still exists with respect to the choice of predictors.

3 Conclusion

We believe that the simplification that the investors' learning problem is linear and investors know the functional form may be not representative of reality. Indeed the complexity of the problem increases in the empirical setting taking into consideration a second-order polynomial but at the same time reducing the set of predictors. Therefore, while in the modeling part they address the gap in the literature of a bounded set of predictors, they still maintain linearity assumptions (the potentially predictable component of future cash-flow growth is linear). On the other hand, in the empirical setting, they do the opposite (reducing J and accounting for a more complex functional form). Therefore, the trade-off between the high dimensional complexity of the problem and the interpretable solution persists (**the missing piece**). To solve this problem we suggest using a neural network algorithm. Indeed, the curse of dimensionality seems to be a case that machine learning may be able to address. In the specific case, the algorithm would be able to deal with an econometrics "overidentification" case ($J > N$) while using nonlinear functions to fit the dataset. The drawbacks regard mainly the interpretability of the algorithm (especially with respect to a possible variable importance frame for the predictors) and the computational complexity. We suggest the adaptation of an algorithm a la Kuriksha (2021) translating the saving problem of agents (see Aiyagari (1994)) into investment choices.

To conclude, we propose future research directions for entrepreneurship and entrepreneurial finance. Shifting from the asset pricing angle, we reconnect the frame studied in class. De facto, a natural extension over an argument closer to the one studied in class would consider the adaptation of the model in a variable importance argument for venture capitalists' decisions. In other words, we may apply a similar model for VC decisions in order to exploit which are the J characteristics that drive VC investment choices. This study may shed light on the investment process and define best policies accounting for behavioral facets, therefore overcoming the issues and limitations of the assumption of the rational expectation framework.

¹To recall the authors' motivation, we report the following citation: "In any case, the purpose of the empirical analysis is not to provide a formal test of the model but rather to illustrate, in a simple setting with a large number of predictors, the wedge between in-sample and out-of-sample predictability".

References

- Aiyagari, S Rao (1994). “Uninsured idiosyncratic risk and aggregate saving”. In: *The Quarterly Journal of Economics* 109.3, pp. 659–684.
- Fama, Eugene F and Kenneth R French (1993). “Common risk factors in the returns on stocks and bonds”. In: *Journal of financial economics* 33.1, pp. 3–56.
- Kuriksha, Artem (2021). “An Economy of Neural Networks: Learning from Heterogeneous Experiences”. In: *arXiv preprint arXiv:2110.11582*.
- Martin, Ian WR and Stefan Nagel (2022). “Market efficiency in the age of big data”. In: *Journal of Financial Economics* 145.1, pp. 154–177.
- Michalski, Lachlan and Rand Kwong Yew Low (2021). “Corporate credit rating feature importance: Does ESG matter?” In: *Available at SSRN 3788037*.

Appendix

A.1 The Zero-Sum Game

Let us simply think of single agents interaction case (no institutional investors). When the outstanding stocks of a firm are all traded on the secondary market, we may consider a simplifying assumption that individuals trade stocks with each other. It is therefore straightforward to state that the gains of an individual equal the losses (or unrealized gains) of another agent. In the trivial case of an agent buying stocks from another individual, if the price of those stocks increases, the agent on the bull side (she bought the stocks) will experience a profit that is exactly equal to the unrealized gain of the "bear" agent (she sold the stock). This result may be easily extended to a more realistic case with N individuals and with other securities in play (think about the derivatives trades as a direct and trivial case).